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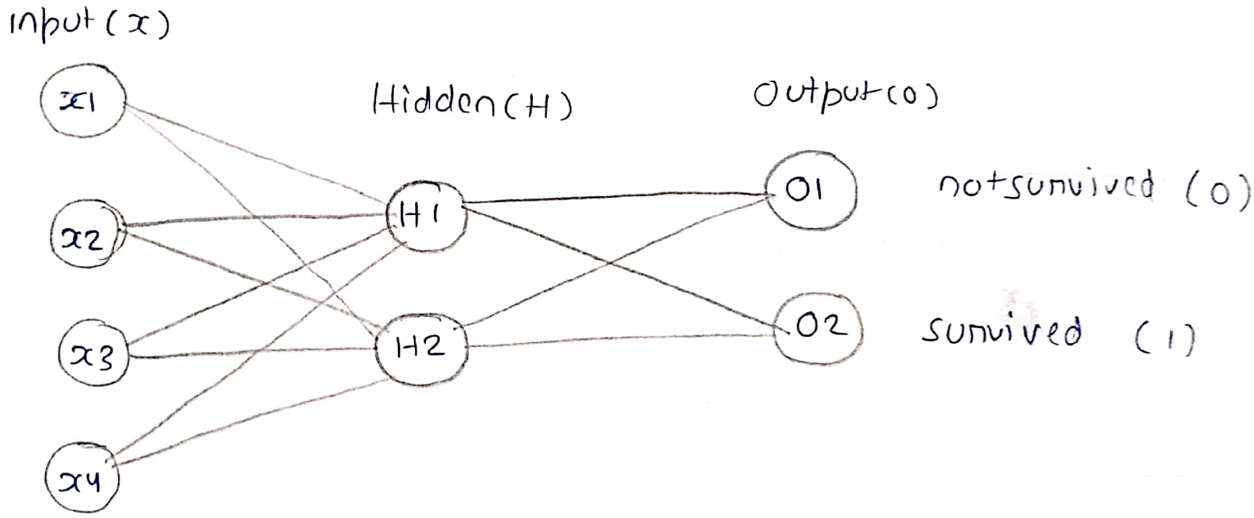
# Assignment

The titanic survival dataset is a classification dataset taken From kaggle it is used to predict whether a passenger survived the disaster or not based on Passengers features such as passenger class age, their fare paid and no of siblings spouse aboard. The dataset has 2 o/p classes 0 not survived and 1 - survived.




input 4
Pclass = 1
Age = 38
Fare = 71.2833
SibSp = 1

O/P 2
no survived
survived.

First row $x = [1, 38, 71.2833, 1]$

Hidden layer weight & Bias.

For H1: $w1 = 0.01, w2 = 0.02, w3 = 0.01, w4 = 0.03$
Bias $B1 = 0.5$
For H2: $w5 = 0.02, w6 = 0.01, w7 = 0.02, w8 = 0.01$
Bias $B2 = 0.3$

Hidden layer H1

$$z1 = x1w1 + x2w2 + x3w3 + x4w4 + B1$$
$$z1 = (1 \times 0.01) + (38 \times 0.02) + (71.2833 \times 0.01) + (1 \times 0.03) + 0.5$$
$$\cancel{z1 = 0.01 + 0.78 + 1.425666 + 0.01 + 0.3}$$
$$z1 = 2.012833$$

Activation (Sigmoid)

$S(z1) = 1/(1+e^{\wedge}-2.012833)$

$H1 = \underline{0.8822}$

Hidden Layer H2

$z2 = x1w5 + x2w6 + x3w7 + x4w8 + B2$

$z2 = (1 \times 0.02) + (38 \times 0.01) + (71.2833 \times 0.02) + (1 \times 0.01) + 0.3$

$z2 = 0.02 + 0.38 + 1.425666 + 0.01 + 0.3$

$z2 = \underline{2.135666}$

Activation (Sigmoid)

$S(z2) = 1/(1+e^{\wedge}-2.135666)$   $H2 = 0.8943.$

Output Layer

O1 -> not survived, O2 -> survived

Let $wH1,O1 = 0.4$, $wH2,O1 = 0.2$, $BO1 = 0.1$

$wH1,O2 = 0.3$, $wH2O2 = 0.8$, $BO2 = 0.1$

For not survived

z(not survived) = H1(0.4) + H2(0.2) + 0.1

$= (0.8822 \times 0.4) + (0.8943 \times 0.2) + 0.1$

$0.35288 + 0.17886 + 0.1$

z(not survived) = $\underline{0.6317}$

For survived

z(survived) = H1(0.3) + H2(0.8) + 0.1

$= (0.8822 \times 0.3) + (0.8943 \times 0.8) + 0.1$

$0.26466 + 0.71544 + 0.1$

z(survived) = $\underline{1.0801}$

output values

[0.6317, 1.0801]

Since $1.0801 > 0.6317$,

The highest output is survived

Predicted output = survived (1)

Scanned with OKEN Scanner